CACD

SEC- 1

Adaptive Decoy-Based Cryptographic Framework with Context-Aware User Behavior Analysis for Insider Threat Defense

Project team members:

ARUGUNTA SHARVANI – 2420090052

NADENDLA PHANINDRA BABU – 2420030553

VARSHA SRITEJA – 2420030331

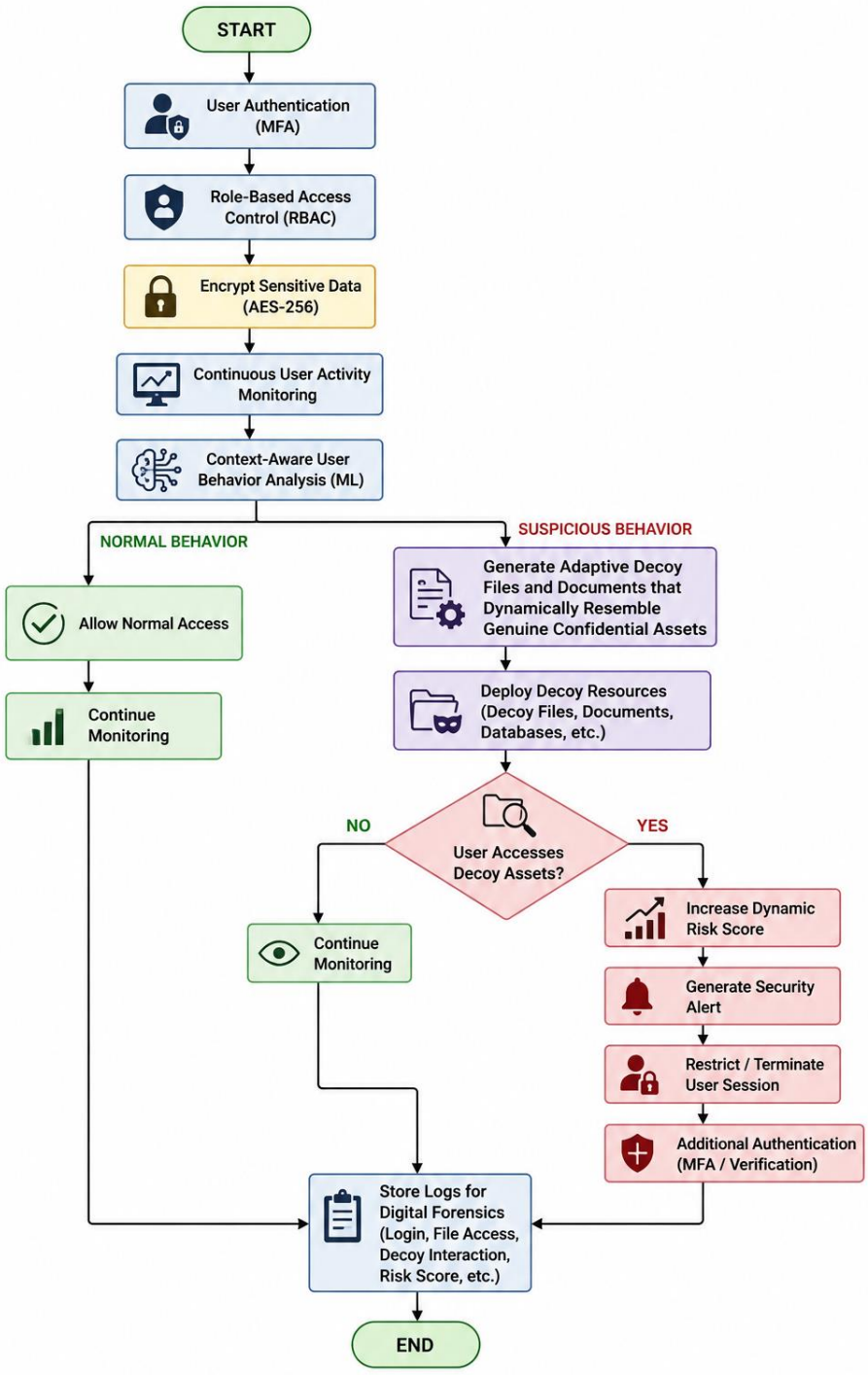
Objectives :

The proposed framework aims to strengthen organizational security against insider threats by integrating adaptive decoy technology, cryptographic protection, and user behavior analytics. The objectives are:

1. To detect malicious insider activities by continuously monitoring user behavior and identifying anomalous actions.
2. To protect sensitive organizational data using advanced cryptographic techniques such as AES-256 encryption.
3. To generate adaptive decoy files and documents that dynamically resemble genuine confidential assets.
4. To lure malicious insiders toward decoy resources and collect evidence of unauthorized access attempts.
5. To perform context-aware behavioral analysis using parameters such as login time, device, location, file access frequency, and privilege level.
6. To calculate a dynamic risk score for every user based on behavioral deviations.
7. To trigger automated responses such as user alerts, account restriction, or session termination when suspicious behavior is detected.
8. To reduce false positives by considering contextual information rather than relying only on predefined security rules.

9. To maintain detailed forensic logs for post-incident investigation.
10. To improve overall insider threat defense with minimal impact on legitimate users.

Flowchart:



Dataset:

For evaluating insider threat detection and user behavior analysis, the CERT Insider Threat Dataset (CMU Software Engineering Institute) is used. It contains synthetic enterprise logs covering normal and malicious user activities such as logins, email, file access, web browsing, and USB usage.

https://kilthub.cmu.edu/articles/dataset/Insider_Threat_Test_Dataset/12841247/1

Methodology overview:

Adaptive Decoy-Based Cryptographic Framework with Context-Aware User Behavior Analysis for Insider Threat Defense is designed to detect and mitigate insider threats by combining secure authentication, cryptographic protection, machine learning-based behavioral analysis, adaptive decoy technology, and automated incident response. The methodology consists of the following phases:

Step 1: User Authentication and Access Control

Every user must authenticate using secure login credentials along with **Multi-Factor Authentication (MFA)**. After successful authentication, **Role-Based Access Control (RBAC)** grants access only to the resources permitted for the user's role and privileges. This ensures that users cannot access unauthorized organizational data.

Step 2: Data Encryption

All confidential organizational data is protected using the **Advanced Encryption Standard (AES-256)**. Sensitive files are encrypted before storage, ensuring confidentiality even if they are copied or stolen. Encryption keys are securely managed and made accessible only to authorized users.

Step 3: Context-Aware User Behavior Monitoring

The system continuously monitors user activities to establish a behavioral profile. Instead of relying solely on predefined security rules, it considers contextual information such as:

- Login time and frequency
- Login location (IP or geographical region)
- Device information
- File access patterns
- USB device usage
- Email communication
- Download and upload activities
- Working hours
- Access to confidential folders

These behavioral features are collected in real time and compared with each user's historical activity.

Step 4: Machine Learning-Based Anomaly Detection

A machine learning model analyzes the collected behavioral data to distinguish between normal and abnormal user activities. The model identifies deviations from established behavioral patterns and computes a **dynamic risk score** for every active user. Users exhibiting suspicious behavior are flagged for further investigation.

Step 5: Adaptive Decoy Generation and Deployment

When suspicious behavior is detected, the framework automatically generates **adaptive decoy files and documents that closely resemble genuine confidential organizational assets**. These decoys are customized according to the user's department, role, ongoing projects, and frequently accessed document types, making them highly convincing. The generated decoys are strategically deployed within accessible directories to attract malicious insiders while remaining invisible to legitimate users.

Step 6: Insider Threat Detection

The system continuously monitors interactions with the deployed decoy assets. Since legitimate users have no operational reason to access these files, any attempt to open, copy, modify, or transfer a decoy is treated as a strong indicator of malicious intent. Such interactions significantly increase the user's risk score and trigger additional verification.

Step 7: Automated Incident Response

Upon confirming suspicious activity, the framework automatically initiates predefined security actions, including:

- Generating real-time alerts for security administrators
- Increasing the user's risk score
- Requesting additional authentication
- Restricting access to sensitive resources
- Temporarily suspending or terminating the active session
- Preserving evidence for further investigation

These automated responses minimize the time between threat detection and mitigation.

Step 8: Logging and Digital Forensics

Every significant event—including authentication records, user activities, encryption operations, behavioral anomalies, decoy interactions, alerts, and response actions—is securely recorded in tamper-resistant audit logs. These logs support digital forensic investigations, compliance auditing, and continuous improvement of the behavioral analysis model.